

Plasma-Assisted Space-Charge Neutralization for High-Current Cyclotron Axial Injection Using Beam-Ionized H₂ and Kr Gas

Abstract

High-current compact cyclotrons are fundamentally constrained by strong space-charge forces during axial injection, where low-energy proton beams experience rapid transverse expansion before entering the spiral inflector. We present a novel plasma-assisted neutralization module designed as a compact, drop-in device installed between the final solenoid and the inflector entrance. The concept employs a short solenoid enclosing a controlled-pressure beamline cell, in which residual H_2 gas or a small admixture of Kr is ionized directly by a 30 keV, multi-milliampere proton beam to form a confined plasma column. Biased end-electrodes trap electrons and accelerate the build-up of space-charge compensation, enabling effective neutralization on sub-0.1~ms timescales. Krypton seeding, with its larger ionization cross section, provides faster transient response and improved stability for pulsed or ramped injection conditions. Initial analytic estimates and particle-in-cell simulations indicate substantial reduction of effective perveance and transverse blow-up, while adding only minimal multiple scattering over the short transport length. The proposed module offers a flexible and compact method to enhance injection efficiency in milliampere-class proton cyclotrons and provides a platform for detailed studies of beam–plasma interaction in strongly space-charge-dominated, low-energy transport. Design considerations, parameter ranges, and predicted performance are presented.